

HW # 29

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CS-F

Problem 1:

$$F_0 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad \lambda's = ? \quad c's = ?$$

$$F_n - F_{n-1} - F_{n-2} = 0$$

$$\lambda^2 - \lambda - 1 = 0$$

$$\lambda = \frac{1 \pm \sqrt{1+4}}{2} = \frac{1 \pm \sqrt{5}}{2}$$

$$\lambda_0 = \frac{1 + \sqrt{5}}{2} \quad \text{or} \quad \lambda_1 = \frac{1 - \sqrt{5}}{2}$$

$$x_n = c_0 \lambda_0^n + c_1 \lambda_1^n + \dots + c_{k-1} \lambda_{k-1}^n$$

let $n=0$ and $n=1$ (initial condition)

$$= c_0 \lambda_0^2 + c_1 \lambda_1^2$$

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ \lambda_0 & \lambda_1 \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \end{bmatrix}$$

$$\left[\begin{array}{cc|c} 1 & 1 & 0 \\ \frac{1+\sqrt{5}}{2} & \frac{1-\sqrt{5}}{2} & 1 \end{array} \right] \sim \left[\begin{array}{cc|c} 1 & 1 & 0 \\ 0 & -\sqrt{5} & 1 \end{array} \right]$$

$$\sim \left[\begin{array}{cc|c} 1 & 0 & +1/\sqrt{5} \\ 0 & 1 & -1/\sqrt{5} \end{array} \right] \Rightarrow \begin{aligned} c_0 &= 1/\sqrt{5} \\ c_1 &= -1/\sqrt{5} \end{aligned}$$

we can find value of x_2 now,

$$x_2 = \frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2} \right)^2 - \frac{1}{\sqrt{5}} \left(\frac{1-\sqrt{5}}{2} \right)^2$$

$$= \frac{\cancel{6+2\sqrt{5}} + \cancel{6+2\sqrt{5}}}{4\sqrt{5}}$$

$$x_2 = \frac{2\sqrt{5} + 2\sqrt{5}}{4\sqrt{5}} = 1$$

$$x_2 = x_0 + x_1$$

$$1 = 1 + 0$$

$$1 = 1$$

Proved.

$$x_n = c_0 \lambda_0^n + c_1 \lambda_1^n + \dots + c_{n-1} \lambda_{n-1}^n$$

where $\lambda_0, \lambda_1, \dots, \lambda_{n-1}$ are eigen values
and $c_0, c_1, \dots, c_{n-1}$ are constants determined by

$$\begin{bmatrix} 1 & 1 \\ \lambda_0 & \lambda_1 \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \end{bmatrix} = \begin{bmatrix} x_0 \\ x_1 \end{bmatrix}$$

$$x_n = \frac{1}{\sqrt{5}} \lambda_0^n + \left(\frac{-1}{\sqrt{5}} \right) \lambda_1^n$$

Problem 2:

$$x_n - 6x_{n-1} + 9x_{n-2} = 0$$

$$\lambda^2 - 6\lambda + 9 = 0$$

$$\lambda = \frac{6 \pm \sqrt{36 - 36}}{2}$$

$$\lambda_2 = \lambda_1 = 3 = \lambda$$

$$x_n = c_1 \lambda^n + c_2 n \lambda^n$$

$$x_0 = c_1 + c_2(0) \quad , \quad x_1 = c_1 3 + c_2 3$$

$$c_1 = 1$$

$$x_1 = 3 + 3c_2$$

$$6 = 3 + 3c_2$$

$$c_2 = 1$$

$$x_n = \lambda^n + n \lambda^n$$

$$x_2 = 3^2 + 2(3)^2$$

$$= 9 + 18$$

$$x_2 = 27$$